Technology Fundamentals

A Text-Based PDF for LangChain and RAG Practice

This sample document contains selectable, machine-readable text. It is designed for testing PDF loading, document splitting, embeddings, and retrieval-augmented generation (RAG) workflows.

1. What is Artificial Intelligence?

Artificial intelligence (AI) is a field of computer science focused on building systems that perform tasks commonly associated with human intelligence. Examples include understanding language, recognizing images, making predictions, and helping people make decisions.

Machine learning is a subset of AI. Instead of writing every rule by hand, developers train a model using data. The model learns patterns from examples and uses those patterns to make predictions on new inputs.

2. What is a Large Language Model?

A large language model (LLM) is a machine learning model trained on large amounts of text. It can generate responses, summarize documents, answer questions, and transform text. LLMs process text as tokens, which may represent words, parts of words, or punctuation.

An LLM does not automatically know the contents of a private PDF. A RAG system can retrieve relevant passages from that PDF and provide them as context when the model generates an answer.

3. Understanding Retrieval-Augmented Generation

Retrieval-augmented generation combines information retrieval with text generation. First, a document is loaded and divided into smaller chunks. Each chunk is converted into a numerical representation called an embedding. The embeddings are stored in a vector database.

When a user asks a question, the system converts the question into an embedding and searches for chunks that are semantically similar. The retrieved chunks are then included in the prompt sent to the language model.

4. Why Document Chunking Matters

A chunk is a smaller piece of a larger document. Chunking helps keep retrieved context focused and manageable. If chunks are too large, retrieval may include irrelevant information. If chunks are too small, useful context may be separated.

A common starting point is a chunk size of around 1,000 characters with an overlap of around 200 characters. These are only starting values. The best settings depend on the document structure, the embedding model, and the questions users are likely to ask.

5. Key Terms

Document: A text unit with content and metadata.
Chunk: A smaller text segment created from a document.
Embedding: A vector representation of text.
Vector database: A system that stores vectors and supports similarity search.
Retriever: A component that finds relevant documents or chunks.

Practice Questions

1. What is the difference between AI and machine learning?
2. Why does a RAG system split documents into chunks?
3. What is an embedding?
4. What happens after a user submits a question to a RAG application?